W6L2

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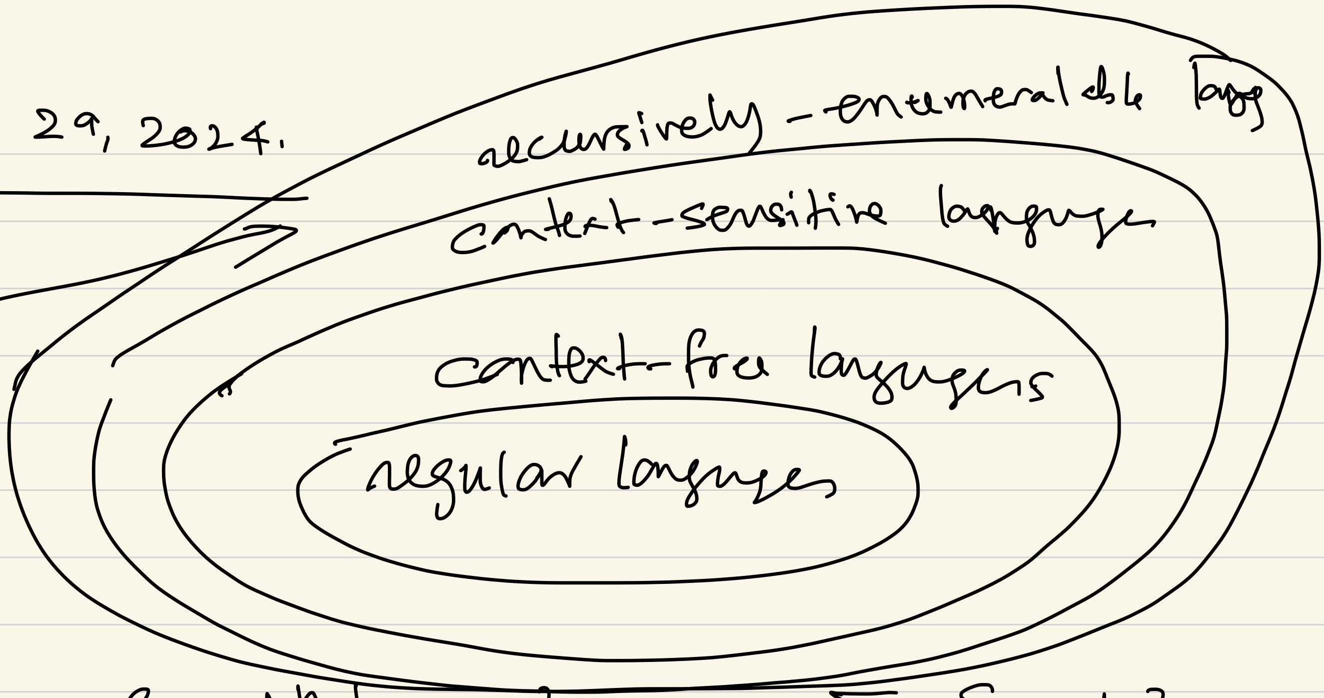

$L_1 = \{a^n b^n \mid n \geq 0\}$ $\Sigma = \{a, b\}$

Turing Machine

$A = \{w\#w\}$ w is over some alphabet Σ.

Suppose $\Sigma = \{0, 1, \#\}$

$\cancel{0}\cancel{1}\cancel{0}\cancel{0}\cancel{1} \# \cancel{0}\cancel{1}\cancel{0}\cancel{0}\cancel{1}$ ___ $\frac{0}{1}$

✓

M L

First possibility.

x_1

$$\xrightarrow{x_1} \boxed{M} \longrightarrow \text{accept}$$

$x_1 \in L$

Second possibility

$$\xrightarrow{x_2} \boxed{M} \longrightarrow \text{reject}$$

$x_2 \notin L$

third possibility

$$\xrightarrow{x_3} \boxed{M}$$

1st case: $x_1 \in L$

2nd case: $x_2 \notin L$

3rd case: $x_3 \overset{\in}{\underset{\notin}{}} L$

A language L for which we can design a TM that goes in either 1st case or 2nd case is called

Turing-Decidable Language.

Turing-recognizable language.

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we have states in a Turing Machine.

λ-calculus